

# Final

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## Github Repository

### Set Up

```
# loading in relevant packages
library(tidyverse)
library(here)
library(janitor)
library(DHARMA)
library(ggeffects)
library(rstatix)

# reading in nest observation data
nests <- read.csv(here("data", "nest_data_final.csv"))
# reading in personal exercise and screen time data
exercise <- read.csv(here("data", "exercise.csv"))
```

### Problems

#### Problem 1. Research Writing

##### a. Transparent Statistical Methods

In part 1, they used a **binomial “log odds” regression** to relate and predict (*Recurvirostra americana*) habitat occupation to distance from the water’s edge.

In part 2, they used a **Chi-Squared test** to compare expected vs observed counts for habitat type and occupation by species.

##### b. Suggestions for Rewriting

Utilizing binomial logistic regression analysis, we found a significant relationship between distance from the water’s edge and American Avocet (*Recurvirostra americana*) microhabitat use.

Utilizing a Chi-Squared test, we found a significant relationship between bird species (American Avocet, Forster’s Tern, Black-necked Stilt) and habitat (managed pond, non-tidal marsh, salt pond, tidal marsh, or tidal mudflat).

### c. Further Analysis

One additional variable that could influence the probability of American Avocet (*Recurvirostra americana*) microhabitat use is **salinity**. This variable is continuous, numerical (ppm) and often calculated after measuring electrical conductivity in micro Siemens per centimeter ( $\mu\text{S}/\text{cm}$ ). Salinity influences a plethora of biological interactions, specifically dehydration, bodyweight and feeding/preening behavior.[\(link\)](#).

## Problem 2. Data Analysis

### a. Clean & Wrangle

```
nests_clean <- nests |>
  # retaining only columns needed for analysis
  select(case_control,
         ant_species,
         height_cm) |>
  # keeping only the three relevant ant species
  filter(ant_species %in% c("AB",
                          "RRB",
                          "BBR")) |>
  # replacing species codes with scientific names
  mutate(ant_name = case_match(
    ant_species,
    "AB" ~ "C.sjostedti",
    "RRB" ~ "C.mimosae",
    "BBR" ~ "C.nigriceps"
  )) |>
  # converting ant_name to a factor and setting display order
  mutate(ant_name = as_factor(ant_name),
         ant_name = fct_relevel(ant_name,
                                "C.sjostedti",
                                "C.mimosae",
                                "C.nigriceps"))

exercise_clean <- exercise |>
  # standardizing column names
  clean_names() |>
  # retaining only columns needed for analysis
  select(screen_time,
```

```

        exercise_time,
        exercise_type)

# checking column types and structure
str(nests_clean)

'data.frame':  270 obs. of  4 variables:
 $ case_control: int  1 0 0 0 0 1 0 0 0 1 ...
 $ ant_species : chr  "RRB" "RRB" "RRB" "RRB" ...
 $ height_cm   : num  392 310 513 154 324 ...
 $ ant_name    : Factor w/ 3 levels "C.sjostedti",...: 2 2 2 2 1 2 3 2 1 3 ...

# previewing 10 random rows
slice_sample(nests_clean, n = 10)

```

|    | case_control | ant_species | height_cm | ant_name    |
|----|--------------|-------------|-----------|-------------|
| 1  | 0            | RRB         | 221.5     | C.mimosae   |
| 2  | 0            | RRB         | 173.5     | C.mimosae   |
| 3  | 0            | RRB         | 163.5     | C.mimosae   |
| 4  | 0            | RRB         | 118.0     | C.mimosae   |
| 5  | 0            | RRB         | 147.0     | C.mimosae   |
| 6  | 0            | RRB         | 153.5     | C.mimosae   |
| 7  | 1            | BBR         | 157.5     | C.nigriceps |
| 8  | 0            | RRB         | 353.0     | C.mimosae   |
| 9  | 0            | RRB         | 232.0     | C.mimosae   |
| 10 | 0            | RRB         | 123.0     | C.mimosae   |

## b. Response Variable

The 1s and 0s present within the “case\_control” column of the data represent if a bird nest was observed (1) or not (0) within a specific tree.

## c. Purpose of Study

**Tree height** is a continuous, numerical variable measured in centimeters. As tree height increases, the probability of nest occurrence would increase because the tree contains more space for nesting and is further away from the ground and therefore potential predators. (Section 3: Results, ¶2)

**Acacia ant species** is a categorical, nominal variable. The four studied ant species differ in aggression of protecting their host trees, as aggressive protection increases, bird nesting also increases because of this inherent nest protection. (Section 3: Results, ¶3)

#### d. Table of Models

| Model number | Predictor List           | Interaction Term? |
|--------------|--------------------------|-------------------|
| 1            | Ant Species, Tree Height | No Interaction    |
| 2            | Ant Species, Tree Height | With Interaction  |

#### e. Running the Models

```
# model 1: additive effects of ant species and tree height on nest presence
model1 <- glm(
  # formula: response: variable + variable
  case_control ~ ant_name + height_cm,
  data = nests_clean,
  # binomial family for binary (0/1) response variable
  family = "binomial")

# model 2: interaction between ant species and tree height on nest presence
model2 <- glm(
  #formula: response ~ variable * interacting variable
  case_control ~ ant_name*height_cm,
  data = nests_clean,
  # binomial family for binary (0/1) response variable
  family = "binomial")
```

#### f. Selecting the Best Model

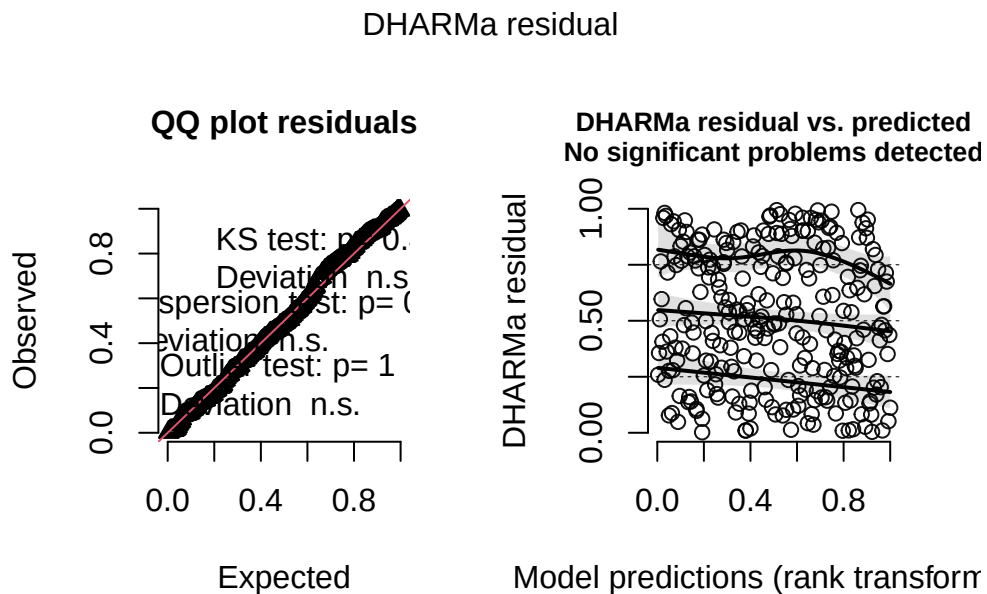
```
# comparing models by AIC; lower AIC indicates better fit
AIC(model1,
  model2)
```

```
      df      AIC
model1  4 258.7430
model2  6 237.8042
```

The best model as determined by Akaike's Information Criterion (AIC) was the model predicting the probability of bird nest occurrence as predicted by ant species interacting with tree height.

## g. Checking Diagnostics

```
# simulating residuals and plotting DHARMA diagnostic plots to check model assumptions
plot(simulateResiduals(model2))
```



## h. Visualizing Model Predictions

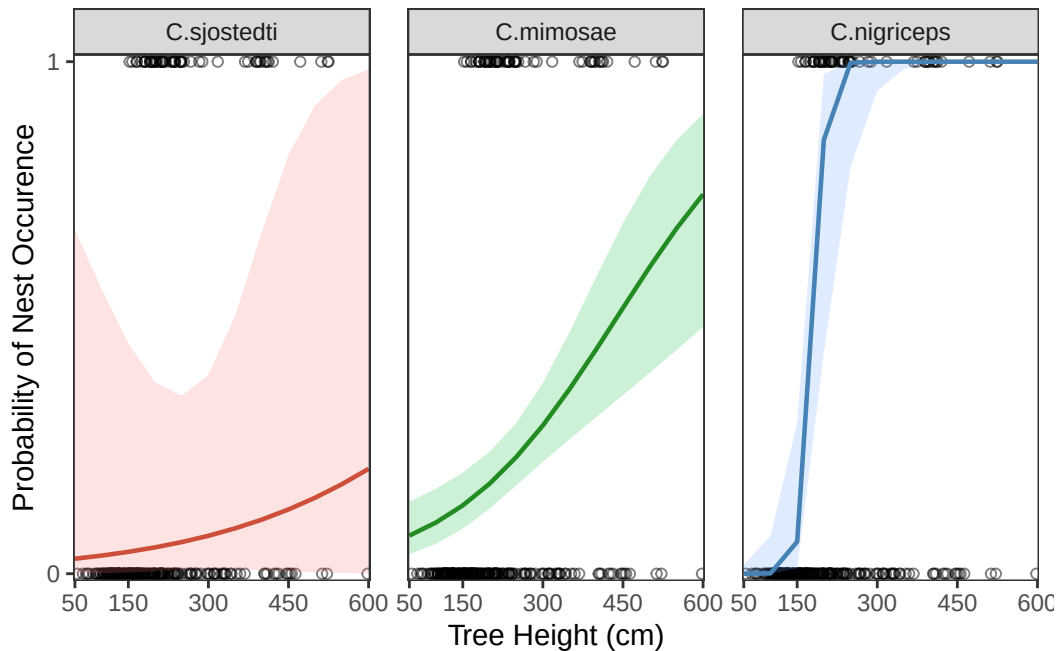
```
# generating predicted probabilities across tree heights for each ant species
mod_preds <- ggpredict(model2, terms = c("height_cm", "ant_name"))

# base layer ggplot, tree height on x-axis, nest ocurence on y-axis.
ggplot(data = nests_clean,
       mapping = aes(x = height_cm,
                     y = case_control)) +
```

```

# plotting raw nest observations as semi-transparent open circles
geom_point(shape = 21,
           alpha = 0.5,
           col = 'black') +
# adding 95% confidence interval ribbons per ant species
geom_ribbon(data = mod_preds,
           aes(x = x,
              y = predicted,
              ymin = conf.low,
              ymax = conf.high,
              fill = group),
           alpha = 0.2) +
# adding model prediction lines per ant species
geom_line(data = mod_preds,
          aes(x = x,
              y = predicted,
              color = group),
          linewidth = 0.8) +
# setting y-axis to span 0-1 probability scale
scale_y_continuous(limits = c(0,1),
                   breaks = c(0,1),
                   expand = c(0.007,0.007)) +
# setting x-axis to observed range of tree heights
scale_x_continuous(limits = c(50,600),
                   breaks = c(50,150,300,450,600),
                   expand = c(0.005,0.005)) +
# non-default theme
theme_bw() +
theme(panel.grid = element_blank(),
      # hiding legend since ant species are labeled via facets
      legend.position = "none",
      panel.spacing.x = unit(0.5,"cm")) +
# labelling axes
labs(x = "Tree Height (cm)",
     y = "Probability of Nest Occurrence") +
# manually assigning colors to each ant species
scale_color_manual(values = c("C.sjostedti" = 'tomato3',
                              "C.mimosae" = 'forestgreen',
                              "C.nigriceps" = 'steelblue')) +
# creating a separate panel per ant species
facet_wrap( ~ group)

```



#### i. Writing a Caption for Figure

**Fig. 2** Ant species and tree height interact to predict probability of bird nest occurrence in whistling thorn trees (*Vachellia drepanolobium*). Black circles represent individual observations of bird nests with values of “1” indicating the presence of a bird nest and “0” indicating no bird nest was observed. Solid, colored lines represent model predictions with shaded ‘ribbons’ representing 95% confidence intervals across ant species. Red represents *C.sjostedti*, Green represents *C.mimosae*, and blue represents *C.nigriceps*. Lujan, E., Nielsen, R., Short, Z., Wicks, S., Nderitu Watetu, W., Malingati Khasoha, L., Palmer, T. M., Goheen, J. R., & Alston, J. M. (2023). Data, code, and metadata for: Symbiotic acacia ants drive nesting behavior by birds in an African savanna [Data set]. Zenodo. <https://doi.org/10.5281/zenodo.8373322>

#### j. Calculating Model Predictions

```
# predicting nest probability at maximum observed height (600 cm) for each ant species
ggpredict(model2, terms = c("height_cm [600]",
                             "ant_name"))
```

```
# Predicted probabilities of case_control
```



ant\_name: *C.sjostedti*

| height_cm | Predicted | 95% CI     |
|-----------|-----------|------------|
| 600       | 0.20      | 0.00, 0.99 |

ant\_name: *C.mimosae*

| height_cm | Predicted | 95% CI     |
|-----------|-----------|------------|
| 600       | 0.74      | 0.48, 0.90 |

ant\_name: *C.nigriceps*

| height_cm | Predicted | 95% CI     |
|-----------|-----------|------------|
| 600       | 1.00      | 1.00, 1.00 |

### k. Interpreting Results

At a tree height of six meters, the probability of nest occurrence is 0.2 (95% CI:[0,0.99]) for *C.sjostedti*, 0.74 (95% CI:[0.48,0.90]) for *C.mimosae*, and 1 (95% CI:[1.00,1.00]) for *C.nigriceps*. As height increases, the probability of bird nesting increases. Additionally, the effect of tree height on bird nest probability is strongest for *C.nigriceps*, followed by *C.mimosae* and weakest for *C.sjostedti*. These values depict the relationship that birds tend to nest in taller trees occupied by *C.nigriceps* and *C.mimosae*, while randomly and less frequently in trees occupied by *C.sjostedti*. This phenomenon is likely due to the acacia ants conferring protection to bird nests, with more aggressive species (*C.nigriceps* and *C.mimosae*) offering better protection from predation.

## Problem 3. Affective & Exploratory Visualizations

### a. Comparing Visualizations

- My exploratory visualizations in HW 2 represents my data through a bar plot and scatter plot with very limited data to show. This contrasts heavily with my visualization in HW 3, which displays much more data and in the form of a jitter plot.

- I see very little similarities between my visualizations as my rudimentary visualizations in HW 2 lacked data. However, I noticed the black and white theme is utilized in both, and I tend to gravitate towards blue as a color for representing my data.
- The relationship between screen time and time spent exercising shifted from a positive linear relationship in HW 2, to a weakly correlated, random relationship in HW 3. The accumulation of data points with 0 time spent exercising displays a large shift in data due to rest days. The difference in screen time *by exercise type* was an interesting pattern that emerged from the data, which was impossible to visualize in HW 2 due to a lack of data.
- I received really good feedback from Professor Bui, regarding a jitter plot as the appropriate form of data visualization, which was implemented throughout further assignments. Furthermore, I received feedback of abstracting screen time into the background of my artist statement from my peers which I found to be a great, creative concept and inputted this advice into my artistic piece.

## b. Designing an Analysis

One analysis that I can run for my data is a Kruskal-Wallis Rank Sum test. My essential research question is how does time spent exercising affect my screen time. However, after more data collection, I have become curious on how the *type of exercise* affects my screen time, does one type of exercise reduce screen time more than another? The Kruskal-Wallis test is appropriate for this question as it will directly compare the median screen time of three different exercise types which are categorical, nominal data.

## c. Checking Assumptions & Running Analysis

Assumption 1: Random Sampling - Because I took the data myself, and recorded independent observations varying by day, each observation is random and non-influential over another data point.

Assumption 2: 5 Minimum Observations Per Group

```
exercise_clean |>
  # counting observations per exercise type to verify minimum sample size assumption
  count(exercise_type)
```

```
exercise_type  n
1           Gym 13
2    Gym + Run  2
```

```
3   No Exercise 10
4           Run   5
```

```
# calculating median screen time per exercise type
aggregate(screen_time ~ exercise_type, data = exercise_clean, FUN = median)
```

```
exercise_type screen_time
1           Gym          181
2   Gym + Run          177
3   No Exercise          182
4           Run          134
```

**Note:** Because the “Gym + Run” group only has two observations, I will omit it from my Kruskal-Wallis Rank Sum test as it violates assumption 2.

```
# running Kruskal-Wallis test, excluding the underpowered "Gym + Run" group
kruskal.test(
  screen_time ~ exercise_type,
  data = exercise_clean,
  subset = exercise_type != "Gym + Run")
```

Kruskal-Wallis rank sum test

```
data: screen_time by exercise_type
Kruskal-Wallis chi-squared = 1.2864, df = 2, p-value = 0.5256
```

```
# post-hoc Dunn test to identify which group pairs differ significantly
exercise_clean |>
  filter(exercise_type != "Gym + Run") |>
  dunn_test(screen_time ~ exercise_type)
```

```
# A tibble: 3 x 9
```

| .y.           | group1      | group2  | n1    | n2    | statistic | p     | p.adj | p.adj.signif |
|---------------|-------------|---------|-------|-------|-----------|-------|-------|--------------|
| * <chr>       | <chr>       | <chr>   | <int> | <int> | <dbl>     | <dbl> | <dbl> | <chr>        |
| 1 screen_time | Gym         | No Exe~ | 13    | 10    | 0.554     | 0.580 | 0.924 | ns           |
| 2 screen_time | Gym         | Run     | 13    | 5     | -0.736    | 0.462 | 0.924 | ns           |
| 3 screen_time | No Exercise | Run     | 10    | 5     | -1.13     | 0.258 | 0.773 | ns           |

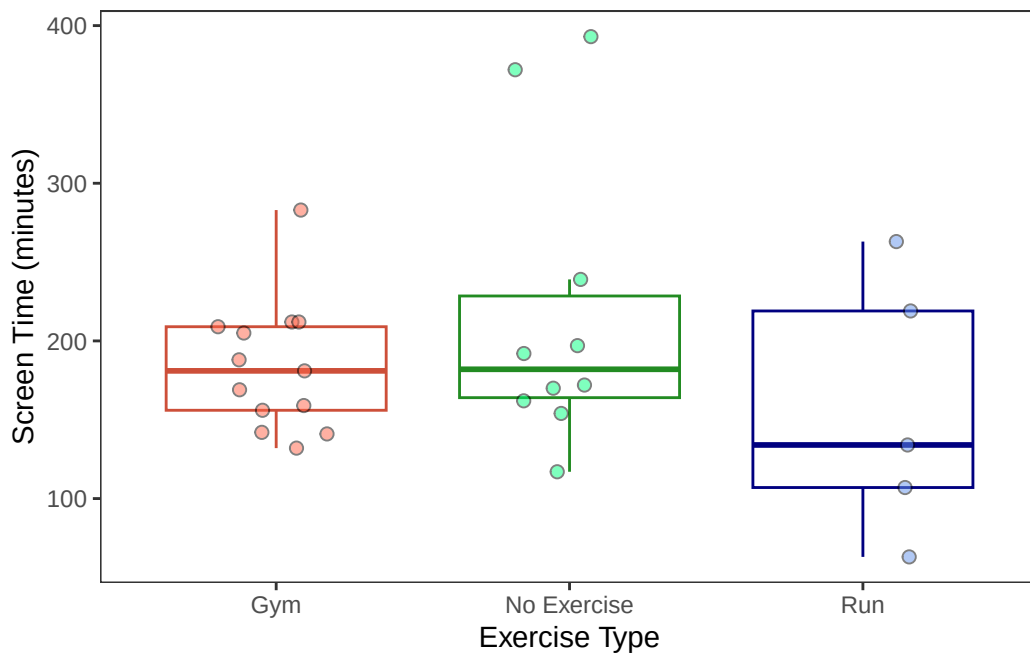
```
# calculating eta-squared effect size for the Kruskal-Wallis result
exercise_clean |>
  filter(exercise_type != "Gym + Run") |>
  kruskal_effsize(screen_time ~ exercise_type)
```

```
# A tibble: 1 x 5
  .y.          n effsize method magnitude
* <chr>      <int>   <dbl> <chr>   <ord>
1 screen_time    28 -0.0285 eta2[H] small
```

#### d. Creating Visualization

```
# using exercise_clean object
exercise_clean |>
  # removing "Gym + Run" rows
  filter(exercise_type != "Gym + Run") |>
  # base layer ggplot
  ggplot(
    # mapping exercise type to x axis and screen time to y axis
    mapping = aes(x = exercise_type,
                  y = screen_time)) +
    # adding boxplots colored by exercise type
    geom_boxplot(aes(color = exercise_type),
                 # removing outliers since we have a jitterplot
                 outlier.shape = NA) +
    geom_jitter(
      # restricting vertical movement to show exact data values
      height = 0,
      # slightly shifting points horizontally to prevent stacking
      width = 0.2,
      # setting point type to a filled circle border
      shape = 21,
      alpha = 0.5,
      size = 2,
      aes(fill = exercise_type)) +
    # manually assigning colors to categories for boxplot
    scale_color_manual(
      values = c("Gym" = "tomato3",
                 "Run" = "navyblue",
                 "No Exercise" = "forestgreen")) +
```

```
# manually assigning colors to categories for jitterplot
scale_fill_manual(
  values = c("Gym" = "tomato",
            "Run" = 'cornflowerblue',
            "No Exercise" = 'springgreen')) +
# non-default theme
theme_bw() +
# removing legend and gridlines
theme(legend.position = "none",
      panel.grid = element_blank()) +
# labelling axes
labs(x = "Exercise Type",
     y = "Screen Time (minutes)")
```



#### e. Writing Caption

**Fig. 3** Daily screen time by exercise type. Circles represent individual observations of screen time in minutes; boxplots represent group medians and interquartile ranges (IQR). Red data denotes resistance training, green denotes no exercise, and blue denotes running.

#### f. Writing About Results

We found a **small** ( $\eta^2 = -0.029$ ) **non-statistically significant** ( $p > \alpha$ ) difference in median screen time between resistance training ( $\tilde{x} = 181$  min,  $n = 13$ ), no exercise ( $\tilde{x} = 182$  min,  $n = 10$ ), and running ( $\tilde{x} = 134$  min,  $n = 5$ ). (Kruskal-Wallis test,  $\chi^2(2) = 1.29, p = 0.53, \alpha = 0.05$ ). Post-hoc Dunn testing further proved no statistically significant difference was present between groups, with the lowest  $p_{adj} = 0.77$ . **Note:** the small ( $n = 5$ ) sample size of the running group may have affected overall statistical outcomes.

#### g. Sharing Affective Visualization

Done!